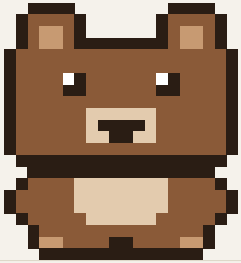
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 2: Causality (II)



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PART 1

Khuzdar and Maria



Notation

In this lecture we use the notation of Angrist and Pischke:

- $Y_{1i} \equiv Y_i(1)$: potential outcome of individual i if treated ($D_i = 1$)
- $Y_{0i} \equiv Y_i(0)$: potential outcome of individual i if not treated ($D_i = 0$)
- Y_i: actual (observed) outcome

Example: Khuzdar and Maria

Two visiting students at MIT have to decide whether to opt in to the university health insurance (HI) plan by paying a fee.

- **Khuzdar** Khalat, from Kazakhstan, has national HI that covers all its citizens. His health is weak to begin with, and he decides to buy the insurance
- **Maria** Moreno, from Chile, is very healthy and little concerned by Boston winters. She does not buy the insurance
- For each individual there are two potential outcomes, whose difference is the causal effect of HI:
  - for Khuzdar: $\tau_K = Y_{1K} - Y_{0K}$
  - for Maria: $\tau_M = Y_{1M} - Y_{0M}$

Example: Khuzdar and Maria

	Khuzdar	Maria
Potential outcome without insurance: Y_{0i}	3	5
Potential outcome with insurance: Y_{1i}	4	5
Treatment (insurance status chosen): D_i	1	0
Actual health outcome: Y_i	4	5
Treatment effect: $Y_{1i} - Y_{0i}$	**1**	**0**

Note: $i = K$ or M; $D_i = 1$ or 0

Example: Khuzdar and Maria

- Only one of the two potential outcomes is revealed
- Khuzdar and Maria make different HI choices, and their actual outcomes Y_i differ:
  - Khuzdar's actual (realised and observed) health outcome is $Y_K = 4$
  - Maria's is $Y_M = 5$
- If we make a **naive** comparison of the insured Khuzdar with the uninsured Maria, we get $Y_K - Y_M = -1$
- From this raw comparison between healthy Maria and frail Khuzdar, we could conclude that getting insurance is counterproductive

Example: Khuzdar and Maria – selection bias

The comparison of actual outcomes does not identify a causal effect. Linking observed and potential outcomes, and adding and subtracting $Y_{0K} = 3$:

$$Y_K - Y_M = Y_{1K} - Y_{0M} = \underbrace{(Y_{1K} - Y_{0K})}_{4-3} + \underbrace{(Y_{0K} - Y_{0M})}_{3-5} = -1$$

We get two hidden comparisons:

- $Y_{1K} - Y_{0K} = 1$: the causal effect of HI on Khuzdar, i.e. τ_K
- $Y_{0K} - Y_{0M} = -2$: the difference in health between Khuzdar and Maria if neither had HI

> The second term shows a lack of comparability between K and M:
> **selection bias**

PART 2

From people to firms



The same problem, for firms: Alfa and Beta

A regional agency offers an **investment grant**: it pays 30% of the cost of new machinery. Two manufacturing SMEs decide whether to apply.

- **Alfa Srl**: well managed, already exporting, with an ambitious investment plan. It applies and gets the grant
- **Beta Srl**: a family firm, busy with day-to-day operations, no one in charge of public calls. It does not apply
- Outcome Y_i: **sales growth** 2015–2019 (%)
- For each firm there are two potential outcomes, whose difference is the causal effect of the grant:
  - for Alfa: $\tau_A = Y_{1A} - Y_{0A}$
  - for Beta: $\tau_B = Y_{1B} - Y_{0B}$

Alfa and Beta

	Alfa	Beta
Potential sales growth without the grant: Y_{0i}	8	3
Potential sales growth with the grant: Y_{1i}	11	6
Treatment (applied and got the grant): D_i	1	0
Actual sales growth: Y_i	11	3
Treatment effect: $Y_{1i} - Y_{0i}$	**3**	**3**

Note: hypothetical numbers; the grant has the same effect, $+3$ points, on both firms

Alfa and Beta – selection bias

The agency compares the two firms it observes:

$$Y_A - Y_B = Y_{1A} - Y_{0B} = \underbrace{(Y_{1A} - Y_{0A})}_{11-8\,=\,3} + \underbrace{(Y_{0A} - Y_{0B})}_{8-3\,=\,5} = 8$$

- $Y_{1A} - Y_{0A} = 3$: the causal effect of the grant on Alfa, τ_A
- $Y_{0A} - Y_{0B} = 5$: how much faster Alfa would have grown than Beta **anyway**, without any grant

> The naive comparison says $+8$; the grant did $+3$. The other 5 points are **selection bias**: the grant takes credit for Alfa's management

Which way does the bias go?

- **Khuzdar and Maria**: the frail student buys insurance $\Rightarrow Y_{0K} < Y_{0M} \Rightarrow$ the naive comparison **understates** the effect (it even gets the sign wrong)
- **Alfa and Beta**: the stronger firm applies $\Rightarrow Y_{0A} > Y_{0B} \Rightarrow$ the naive comparison **overstates** the effect

> The sign of the selection bias depends on **who** selects into treatment, and why

Who selects into a firm subsidy?

For firms, selection happens twice:

- **self-selection**: which firms know about the call, can write a project, can co-finance the investment
- **selection by the agency**: applications are ranked and the best projects are funded (as in Law 488/1992, week 5): "picking winners"

> **YOUR TURN**
>
> Can you think of a policy where the **weakest** firms select into treatment? (Hint: rescue aid, short-time work schemes, debt moratoria.) Which way does the bias go then?

PART 3

Subsidised firms



Case: an investment grant for manufacturing SMEs

- Data: **1,500 Italian manufacturing SMEs**, observed in 2015 (before the call) and 2019
  - in 2016 a regional agency funds investment projects, in the spirit of Law 488/1992
  - the data are **simulated** for this course: since we generated them, we know the true effect, and we will check our methods against it
- **Outcome** Y_i: sales growth 2015–2019 (%)
- **Treatment** D_i: the firm received the grant
  - treatment group: subsidised firms, $D_i = 1$ (419 firms)
  - control group: non-subsidised firms, $D_i = 0$ (1,081 firms)

> The dataset, `firms_grant.csv`, is the one we open in R in Lab 1 on Friday: you will reproduce every number in this lecture

Table 1: outcomes of subsidised and non-subsidised firms

	Grant (1)	No grant (2)	Difference (3)
Sales growth 2015–19 (%)	9.76	3.37	6.39
	[9.04]	[8.66]	(0.51)
Employment growth 2015–19 (%)	4.54	1.67	2.86
	[7.76]	[7.48]	(0.44)
Number of firms	419	1,081	

Standard deviations in brackets; standard error of the difference in parentheses. Simulated data, `firms_grant.csv`.

- A simple comparison of averages suggests that the grant works: subsidised firms grew 6.4 percentage points faster

Causal effect?

- Are the sales of a subsidised firm growing faster than the sales of the **same** firm would have grown without the grant?
- Looking at the data, it seems so
- However, the result may be misleading: if the agency funds firms like Alfa and not firms like Beta, we are comparing apples with oranges

> Remember: a good **control group** should reveal the fate of the treated in the counterfactual world of no treatment

Balance check

- We need to analyse the two groups more deeply, with the help of statistical tools

→ perform a **balance check**:

  - verify whether there are statistically significant differences in observable characteristics **measured before the treatment** between the treated and the control group

- A **balanced** sample implies that the characteristics of the firms in the treated and in the control group have the same mean

Test of statistical significance: t-test

- Using the sample averages, we compute the **t-statistic**: the difference in means divided by the estimated standard error of this difference

$$t_{stat} = \frac{\bar{X}^1 - \bar{X}^0 - 0}{\widehat{s.e.}(\bar{X}^1 - \bar{X}^0)}$$

- t_{stat} is compared with the critical values of Student's t distribution
- We apply the same test to check whether a coefficient from an estimation is statistically significant, i.e. whether we can reject that the effect is zero:

$$t_{stat} = \frac{coeff - 0}{s.e.(coeff)}$$

compared with the same critical values

T-tests: critical values

- if $|t_{stat}| > 2.58$, the difference is **statistically significant at 1%**, usually indicated with ***
- if $|t_{stat}| > 1.96$, the difference is **statistically significant at 5%**, usually indicated with **
- if $|t_{stat}| > 1.64$, the difference is **statistically significant at 10%**, usually indicated with *

If the **p-value** is provided, compare it with the significance levels: e.g. p-value < 0.01 means significance at 1%.

> Significant at 1%: if the true difference were zero, a difference this large would arise by chance less than 1% of the time

Table 2: characteristics of subsidised and non-subsidised firms

	Grant (1)	No grant (2)	Difference (3)
A. Outcome			
Sales growth 2015–19 (%)	9.76	3.37	6.39 (0.51)
B. Characteristics in 2015, before the call			
Age (years)	23.63	24.00	−0.37 (0.68)
Employees	33.09	24.65	8.44 (1.38)
Sales (thousand EUR)	7,257	4,387	2,870 (393)
Exporter (share)	0.47	0.23	0.24 (0.03)
Family-owned (share)	0.71	0.77	−0.05 (0.03)
Located in the South (share)	0.44	0.30	0.15 (0.03)
Number of firms	419	1,081	

Standard errors of the differences in parentheses. Simulated data, `firms_grant.csv`.

Balance check: two t-tests

Employees

- $\bar{X}^1 - \bar{X}^0 = 33.09 - 24.65 = 8.44$
- $t_{stat} = \frac{8.44-0}{1.38} = 6.13 > 2.58$
- Significant at 1%

> Subsidised firms are **larger**

Age

- $\bar{X}^1 - \bar{X}^0 = 23.63 - 24.00 = -0.37$
- $t_{stat} = \frac{-0.37-0}{0.68} = -0.54$, and $|-0.54| < 1.64$
- Not significant, even at 10%

> The two groups have the **same** average age

Subsidised vs non-subsidised: characteristics

- Panel B of Table 2 shows significant differences (col. 3): subsidised firms are on average larger, sell more, are more often exporters and more often located in the South
- Several of these variables are correlated with both growth (Y) and the grant (D)
- Exporters grow faster **and** are more likely to apply for (and win) a grant
- Therefore, the difference in sales growth (6.39 points, significant at 1%) partly reflects the different composition of the two groups

> Non-subsidised firms are likely **not** a good control group for subsidised firms: can we estimate a causal effect?

Causal effect?

- In panel A of Table 2, we see the average Y_{1i} only for subsidised firms and the average Y_{0i} only for non-subsidised firms
- This corresponds to the **difference in means** by treatment status
- Even if the grant had no effect at all ($\tau = 0$), subsidised firms would grow faster, because they are larger, more export-oriented and, as we will see, better managed
- This leads to a **biased** estimate of the effect of the grant on growth

> Difference in group means $=$ Average causal effect $+$ Selection bias

PART 4

Selection bias



Average causal effect (I)

The investigation of an average causal effect begins with the comparison of the average sales growth of subsidised and non-subsidised firms.

We can only observe the actual outcome in the two subsamples of size n_1 and n_0:

- $Avg_n[Y_i \mid D_i = 1] = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_{1i}$ among subsidised firms, which is necessarily an average of Y_{1i}, with no information on Y_{0i}
- $Avg_n[Y_i \mid D_i = 0] = \frac{1}{n_0} \sum_{i=1}^{n_0} Y_{0i}$ among non-subsidised firms, which is necessarily an average of Y_{0i}, with no information on Y_{1i}

Average causal effect (II)

In Table 2 we see the average Y_{1i} only for subsidised firms and the average Y_{0i} only for the others. This corresponds to the **difference in means** by treatment status:

$$Avg_n[Y_i \mid D_i = 1] - Avg_n[Y_i \mid D_i = 0] = Avg_n[Y_{1i} \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0] \quad (1)$$

- We are after an average causal effect over every firm's Y_{1i} and Y_{0i},
- but we only see the average Y_{1i} for subsidised firms and the average Y_{0i} for the others

> This leads to a **biased** estimate of the effect of the grant

Constant-effects assumption

Assume that the grant raises the growth of every firm by the same amount τ (the true causal effect):

$$Y_{1i} - Y_{0i} = \tau \quad \Longleftrightarrow \quad Y_{1i} = Y_{0i} + \tau$$

Substituting in (1):

$$\begin{aligned} Avg_n[Y_{1i} \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0] &= Avg_n[Y_{0i} + \tau \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0] \\ &= \tau + \{Avg_n[Y_{0i} \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0]\} \end{aligned}$$

The comparison of subsidised and non-subsidised firms equals the causal effect τ plus the difference in average Y_{0i}: how differently the two groups would have grown **without** the grant.

> Difference in group means $=$ Average causal effect $+$ **Selection bias**

Selection on observables

- When the only source of selection bias is a set of differences in **observable** characteristics, the identification problem is easily fixed
- We can compare firms with the same characteristics
- E.g. if the only source of bias is export status, we can compare subsidised and non-subsidised **exporters** only

In our data:

- all firms: difference in sales growth $= 6.39$ points
- exporters only: 6.27 points; firms with 10–49 employees only: 5.68 points

> Holding one characteristic fixed barely moves the estimate: something else is going on

Selection on unobservables

- With many observable differences (size, exports, location), as in Table 2, it is reasonable to expect **unobserved** differences too
- What makes Alfa apply is rarely in the balance sheet: **management quality**, the entrepreneur's ambition, a good relationship with the bank, someone who reads the calls for funding
- The same features also make Alfa grow faster, grant or no grant

> The main problem in policy evaluation is eliminating the selection bias arising from **unobserved differences**

Our simulated firms: the answer

Because we generated the data, we know the truth:

- true effect of the grant on sales growth: $\tau = $ **3 percentage points** for every firm
- the naive comparison gives **6.39**: more than twice the true effect
- selection bias ≈ 3.4 points, driven mostly by **management quality**, which is not in the dataset

> **LESSON**
>
> An evaluation that compares subsidised with non-subsidised firms would tell the agency the grant works twice as well as it does: money would be shifted towards a policy on the basis of a statistical illusion

PART 5

Inside the firm



The same problem inside the firm

Managers face selection bias too, whenever they evaluate their own decisions.

- A retail chain offers a training course to its store managers; participation is voluntary
- A year later, stores whose manager attended sell 12% more
- Did the course work, or did the most motivated managers sign up?

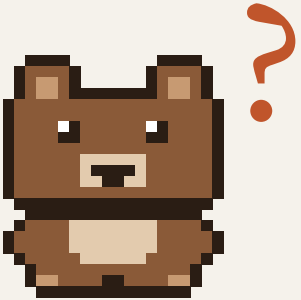
$$\underbrace{\text{Sales gap}}_{\text{observed}} = \underbrace{\text{Effect of training}}_{\text{what HR wants to know}} + \underbrace{\text{Selection bias}}_{\text{who signed up}}$$

COMING UP

Week 3: how firms use pilots and A/B tests to get rid of the second term

Next steps

- **Thursday**: Lecture 3, regression. Can we "control for" the differences in Table 2? (Spoiler: only for the ones we observe)
- **Friday**: **Lab 1**, our first session in R
  - bring a laptop with **R** and **RStudio** installed: https://posit.co/download/rstudio-desktop
  - download `firms_grant.csv` from the course website
  - we will reproduce Tables 1 and 2 of today's lecture



Questions?

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